DAY 01

< UNIX ENVIRONMENTS />

DAY 01

First things first - Intranet & Manifesto

Using the intranet, register for all of the week's activities.

Now, take a moment to read the document given alongside this one : ”the manifesto”, to understand **our values**.

Task 00 - Before we start...

Before diving into discovering your environment, we would like you to do one thing :

Join the **official Discord server** of Epitech to get in touch with all other students and national staff members.

> When you join the Discord server, go to the `#commands` channel and type `/login` to link your Discord account with your Epitech account.

Linux - Discovering your environment

First, take some time to discover your working environment (keywords, commands,...)
Then learn how to lock your workstation, it might prove very useful.

Task 01 - Create a delivery directory

Create a directory named `delivery` in your home directory's root.
Within that directory, create another one named `task01`, which will contain:

- ✓ an empty file named `test01` with the default permissions,
- ✓ a file named `test02` containing "**If you don't struggle, you don't improve.**". Everyone must have the rights to read and execute this file, but only you can write in it,
- ✓ a symbolic link (symlink) named `test03`, which is a reference to `test02`.

Read the **ln** man carefully!

Task 02 - Z

Create a file `delivery/task02/z`, which displays the character '**Z**' followed by a line feed (#cr) when the binary `cat` is used to read it:

```
Terminal
$>  cat -e z
Z$
```

Task 03 - midLS

Write in `delivery/task03/midLS`, a command that lists the current repository's files and directories (without hidden files, ellipses, or files starting with a dot) sorted alphabetically.
Files and directories should be separated by commas and directories must end with a slash.

Add execution permission to everyone

Task 04 - Access your repositories - git and github.com

From this point onwards, all your projects will have to be turned-in in a Github repositories.

Github repositories are simply **git repositories** hosted on github.com.

So first, create an account on github.com using your Epitech email address.

> If you already have a GitHub account, you can simply add your Epitech's email address to the list of emails linked to your account.

On GitHub, all your Epitech's projects will be grouped in an organization.
For this first time, you're even now part of a team of this organization which give you access to the repository to delivery today's work.

> From now on, for every project, you will be given access to a group on the GitHub organization and thus have access to the project's dedicated repository.

Now generate an ssh key named `id_rsa` and add it to yout GitHub account. This key will allow you to authenticate with github (and thus accessing your repositories) via your terminal.

One last thing: be sure to authorize the previously added ssh key for use with single sign-on on GitHub.com.

> For more information, see the document ”How to turn-in using Github” on the intranet (here).

Don't forget to setup your git configuration so github can identify youself:

Terminal

```
$>  git config --global user.name "Firstname Lastname"
$> git config --global user.email "firstname.lastname@epitech.eu"
```

Do a delivery - Clone a repository and push your work

Now, it is time to check if the previous part really works. To do so, we are going to clone the repository for todays tasks.
The repository is available for git at the following address:
git@github.com:`<$PROMOTIONTEAM>`**/**`<$MODULECODE>`**-**`<$INSTANCECODE>`**-**`<$PROJECTNAME>`**-**`<$GROUPID>`**.git**, where:

- ✓ `<$PROMOTIONTEAM>` is the github team corresponding to your promotion,
- ✓ `<$MODULECODE>` is the code of the project's module,
- ✓ `<$INSTANCECODE>` is the code corresponding to your current semester,
- ✓ `<$PROJECTNAME>` is the project's name,
- ✓ `<$GROUPID>` is the id of your group for this project.

You can check on the organization page of GitHub to see the list of all the repositories you have access to.
From a repository webpage, you can click on the ”Code” button to have access to the repository link (to clone it using SSH).

If it worked, you should have a result similar to:

Terminal

```
$>  git clone git@github.com:EpitechMyPromo-2042/G-CPE-100-NCE-1-1-cpoolday01-1.git
Cloning into 'G-CPE-100-NCE-1-1-cpoolday01-1'...
warning: You appear to have cloned an empty repository.
```

Now, You have to move all task directories (previously created) into the repository of the day.
Then, you have to inform ”git” of your wish to **add** these files.
After that, you have to **commit** a new local revision which contains all these modifications.
Finally, you have to **push** this revision to the remote server.
From now on, commit and push at the end of each task.

Make sure that the file permissions have not been altered during the copy/move.

Have you read the document on the intranet already?

Task 05 - mr_clean

Write a file named `mr_clean` - stored at the root of the repository of the day - a command to find and delete every file from the current directory and all subdirectories which end by `~` or which start **and** end by `#`.
Give execution rights to the owner of the file.

> ⚠️ Only one command is allowed (no ";", neither "&&" or anything...)

> 📢 Read carefully the manual of `find`

Task 06 - push_that.sh

Write a shell script (using bash as shell) named `push_that.sh` at the root of your repository.
The script must add all of the current folder's files and push them to the repository.

The script must take a commit message as parameter and should be able to handle simple problems and still push your files.

> 📢 Having trouble figuring out how to retrieve parameters? `man bash`.

> ⚠️ Where you're writing a script, dont't forget the **shebang**!
> Don't forget to give execution right to shell scripts as well!

> ⚠️ Don't use this script if you share a repository with other people. It's not good practice.

Task 07 - Tree

Reproduce the folders structure displayed below.

Terminal

```
$>  tree -FQ task07 | head -n 32
"task07"/
|-- "1910s"/
|   |-- "1911" -> "../Solvay Conferences on Physics/The theory of radiation and quanta"/
|   `-- "1913" -> "../Solvay Conferences on Physics/The structure of matter"/
|-- "1920s"/
|   |-- "1921" -> "../Solvay Conferences on Physics/Atoms and electrons"/
|   |-- "1924" -> "../Solvay Conferences on Physics/Electric conductivity of metals and
    related problems"/
|   `-- "1927" -> "../Solvay Conferences on Physics/Electrons and photons"/
|-- "1930s"/
|   |-- "1930" -> "../Solvay Conferences on Physics/Magnetism"/
|   |-- "1931" -> "../Solvay Conferences on Chemistry/Constitution and Configuration of
    Organic Molecules"/
|   |-- "1934" -> "../Solvay Conferences on Chemistry/Oxygen, and its chemical and biological
     reactions"/
|   `-- "1937" -> "../Solvay Conferences on Chemistry/Vitamins and Hormones"/
|-- "1940s"/
|   `-- "1947" -> "../Solvay Conferences on Chemistry/Isotopes"/
|-- "1950s"/
|-- "Professors"/
|   |-- "Frederic Swarts"
|   |-- "Hendrik Lorentz"
|   |-- "Paul Karrer"
|   |-- "Paul Langevin"
|   `-- "William Jackson Pope"
|-- "Solvay Conferences on Chemistry"/
|   |-- "Constitution and Configuration of Organic Molecules"/
|   |   `-- "chair" -> "../../Professors/William Jackson Pope"
|   |-- "Isotopes"/
|   |   |-- "chair" -> "../../Professors/Paul Karrer"
|   |   `-- "participants"/
|   |-- "Oxygen, and its chemical and biological reactions"/
|   |   `-- "chair" -> "../../Professors/William Jackson Pope"
|   `-- "Vitamins and Hormones"/
|       `-- "chair" -> "../../Professors/Frederic Swarts"
```

Terminal

```
$> tree -FQ task07 | tail -n 24
`-- "Solvay Conferences on Physics"/
    |-- "Atoms and electrons"/
    |   `-- "chair" -> "../../Professors/Hendrik Lorentz"
    |-- "Electric conductivity of metals and related problems"/
    |   `-- "chair" -> "../../Professors/Hendrik Lorentz"
    |-- "Electrons and photons"/
    |   |-- "chair" -> "../../Professors/Hendrik Lorentz"
    |   `-- "participants"/
    |       |-- "A. Einstein"
    |       |-- "E. Schrodinger"
    |       |-- "H.A. Lorentz"
    |       |-- "M. Planck"
    |       |-- "M. Sklodowska-Curie"
    |       |-- "N. Bohr"
    |       |-- "W. Heisenberg"
    |       `-- "W.L. Bragg"
    |-- "Magnetism"/
    |   `-- "chair" -> "../../Professors/Paul Langevin"
    |-- "The structure of matter"/
    |   `-- "chair" -> "../../Professors/Hendrik Lorentz"
    `-- "The theory of radiation and quanta"/
        `-- "chair" -> "../../Professors/Hendrik Lorentz"

31 directories, 23 files
```

Git handles empty directories differently.

Task 08 - Tar

Create a compressed (with Gzip) tarball of the content of the previous task's directory.

Delivery: `task08/task07.tgz`

man tar

v 5.1

{EPITECH}